Assignment #6 - Milestones, Timeline, and Effort Matrix

For this assignment your project team will produce a list of milestones and two tables indicating your project timeline and individual efforts on specified tasks.

Milestones List:

Your group will provide a listing of planned major milestones and completed project deliverables in the development of your project. Provide some detailed explanation for each of these milestones and deliverables.

Project Milestones: Game Development

This document outlines the key milestones for the game development project, from initial prototyping to final testing and deployment.

1. **Functional Actor Prototypes**
    - **Objective:** Develop basic, functional prototypes for actors.
    - **Deliverables:**
        - **Basic Movement and Behavior Logic:** Implement fundamental locomotion (e.g., walking, running, jumping) and core behaviors.
        - **Collision Detection:** Implement basic optimized collision detection for actors with an environment.
2. **Integration of Game Data with Learning Algorithm**
    - **Objective:** Establish a robust pipeline for data exchange between the game engine and a learning algorithm, allowing the algorithm to influence character behavior and for game data to inform the algorithm's learning.
    - **Deliverables:**
        - **Algorithm Integration:** Successfully integrate the learning algorithm into the game's architecture, ensuring it can receive game state information and output updated character behavior parameters.
        - **Dynamic Character Behavior:** Implement the mechanism by which the learning algorithm actively modifies and influences character actions and decision-making within the game.
        - **Data Storage and Utilization:** Design and implement a system for securely storing relevant game data (e.g., player actions, character performance, environmental interactions) and ensuring this data is effectively fed into and utilized by the learning algorithm for continuous improvement and adaptation. This includes defining data schemas and

logging protocols.

3. **Functional Tool Prototypes and Basic Levels**
    - **Objective:** Develop initial prototypes for all in-game tools and construct foundational levels to test core gameplay mechanics.
    - **Deliverables:**
        - **Tool Functionality:** Implement the core functionality for each in-game tool (e.g., flags). This includes user input, visual feedback, and their intended effects on the game world.
        - **Basic Level Design:** Create a series of simple, unpolished levels designed to demonstrate and test the functionality of characters, tools, and the integrated learning algorithm in a controlled environment. These levels will focus on showcasing individual mechanics rather than complex narrative.
        - **User Interface (UI) Prototypes (for Tools):** Develop basic UI elements for selecting and using tools.
4. **Bug Fixing and Tool Finalization**
    - **Objective:** Address all identified bugs related to character and tool functionality and finalize the design and implementation of all in-game tools.
    - **Deliverables:**
        - **Comprehensive Bug Resolution:** Identify, prioritize, and fix all reported bugs pertaining to character movement, behavior, tool functionality, and the learning algorithm's integration.
        - **Tool Refinement:** Iterate on the design and implementation of all tools based on internal testing and feedback, ensuring they are intuitive, effective, and free of glitches. This includes polishing visual effects and animations associated with tool usage.
        - **Performance Optimization (Initial Pass):** Conduct initial performance profiling for character and tool-related systems and implement basic optimizations to ensure smooth gameplay.
5. **Level Construction**
    - **Objective:** Develop and populate all game levels, incorporating environmental details, gameplay challenges, and narrative elements.
    - **Deliverables:**
        - **Complete Level Design:** Design and build all intended game levels, including layout, environmental assets, lighting, and interactive elements.
        - **Gameplay Encounters:** Integrate and balance various gameplay encounters, puzzles, and challenges within each level, utilizing the finalized characters and tools.
        - **Narrative Integration:** Weave in narrative elements and lore through environmental storytelling, dialogues, or in-game text.
        - **Art and Asset Integration:** Populate levels with finalized art assets, textures, and sound effects to achieve the desired aesthetic and atmosphere.
6. **Testing and Bug Fixing (Comprehensive)**

- **Objective:** Conduct extensive testing across all game systems and levels to identify, report, and resolve all remaining bugs and polish the overall player experience.
- **Deliverables:**
  - **Quality Assurance (QA) Testing:** Perform thorough QA testing, including functional testing, performance testing, compatibility testing, and usability testing, across all platforms.
  - **Bug Tracking and Resolution:** Implement a robust bug tracking system Here's the updated timeline, including start and end dates for tasks from Assignment 5 and milestone completion dates:and ensure all reported bugs are prioritized, reproduced, and resolved efficiently.
  - **Performance Optimization (Final Pass):** Conduct a final round of performance profiling and optimization to ensure the game runs smoothly on target hardware.
  - **Balancing and Tuning:** Fine-tune gameplay mechanics, difficulty curves, and economic systems based on testing feedback.
  - **User Interface/User Experience (UI/UX) Refinement:** Iterate on UI/UX elements based on usability testing to ensure an intuitive and enjoyable player experience.

Table 1. Timeline:

Your group will list all tasks associated with your project (from task list assignment) and indicate the range of dates (start and completion) that this task will be worked upon. Within the table you should include planned completion dates of all milestones indicated in your Milestones List.

Task	Team Member(s)	Start	End
Functional Actor Prototypes	Nathaniel Hafely	10/8/2025	10/13/2025
Develop system for spawning actors	Nathaniel Hafely	10/8/2025	10/13/2025
Develop randomized actor movement and behaviors	Nathaniel Hafely	10/8/2025	10/13/2025

Design and implement a main menu and level select screen	Nathaniel Hafely	10/8/2025	10/13/2025
Develop system to display actor representatives	Nathaniel Hafely	10/8/2025	10/10/2025
Integration of game data with learning algorithm	Andrew Towns …	10/13/2025	10/20/2025
Research which algorithms would be best suited to our use case	Andrew Towns	10/13/2025	10/15/2025
Implement different ML algorithms for training	Andrew Towns	10/15/2025	10/18/2025
Develop cost function logic that is influenced by the tools used in the game	Andrew Towns	10/18/2025	10/20/2025
Export weights to game logic to be used by in-game actors/characters	Andrew Towns	10/18/2025	10/20/2025
Develop Data log and storage system	Shawn Lasrado	10/13/2025	10/16/2025
Build an analytics pipeline to refine raw data	Shawn Lasrado	10/16/2025	10/19/2025
Design data relevancy structure	Shawn Lasrado	10/19/2025	10/20/2025
Functional Tool prototypes with	Nathaniel Hafel…	10/13/2025	10/31/2025

basic levels			
Develop a variety of reinforcement tools for the player to interact with the game	Nathaniel Hafely	10/13/2025	10/31/2025
Bug fixing and Tool finalization	Nathaniel Hafel…	10/31/2025	11/15/2025
Level construction	Nathaniel Hafely	11/15/2025	12/10/2025
Develop and design puzzle levels that force the player to use the reinforcement toolbox to find creative solutions	Nathaniel Hafely	11/15/2025	12/10/2025
Comprehensive testing and bug fixing	Nathaniel Hafel…	12/10/2025	12/20/2025

Table 2. Effort Matrix:

Your group will produce a table indicating a breakdown of how much effort each group member will apply to each task. You can provide percentage of effort or hours of effort; the goal is to create a table that meaningfully represents your expected division of labor for all tasks. As indicated in your task list, there should be a team member of primary responsibility for each task.

Task	Nathaniel Hafely	Andrew Towns	Shawn Lasrado
Functional Actor Prototypes	100%		

Develop system for spawning actors	100%		
Design and implement a main menu and level select screen	100%		
Develop randomized actor movement and behaviors	100%		
Develop system to display actor representatives	100%		
Integration of game data with learning algorithm		50%	50%
Research which algorithms would be best suited to our use case		100%	
Implement different ML algorithms for training		100%	
Develop cost function logic that is influenced by the tools used in the game		100%	
Export weights to game logic to be used by in-game actors/characters		100%	
Develop Data log and storage system			100%
Build an analytics pipeline to refine raw			100%

data			
Design data relevancy structure			100%
Functional Tool prototypes with basic levels	50%	50%	
Develop a variety of reinforcement tools for the player to interact with the game	50%	50%	
Bug fixing and Tool finalization	33.33%	33.33%	33.33%
Level construction	50%	30%	20%
Develop and design puzzle levels that force the player to use the reinforcement toolbox to find creative solutions	50%	30%	20%
Comprehensive testing and bug fixing	33.33%	33.33%	33.33%